Supplementary Materials

This appendix collects the granular tables and figures extracted from the main text to keep the narrative focused on the taxonomy, its empirical impact and its validation. Each is reproducible from the released parquet and the audit scripts.

Table S1: A4 sensitivity sweep on the released v1.0.1 parquet (46,307 stations, 123 systems). Each row reports the absolute A4 count and four invariance metrics against the $\sigma = 3$ reference: Kendall's τ on the rank order of the Top-10 cities (**T10c**) and Top-10 operators (**T10o**) by certified docked-bike count, plus Jaccard similarity on the Top-10 and Top-50 sets (**J10c/J50c, J10o/J50o**). Top-10 city membership is invariant ($J = 1.000$) across the grid; $\tau \geq 0.956$ throughout.

σ	A4 st.	A4 sys.	Share	τ T10c	τ T10o	J10c	J50c	J10o	J50o
2.0	597	89	1.29 %	0.956	0.956	1.000	1.000	1.000	1.000
2.5	543	88	1.17 %	0.956	0.956	1.000	1.000	1.000	1.000
3.0	**500**	**88**	**1.08 %**	ref.	ref.	ref.	ref.	ref.	ref.
3.5	459	82	0.99 %	1.000	1.000	1.000	0.961	1.000	1.000
4.0	415	80	0.90 %	1.000	0.956	1.000	0.961	1.000	1.000

Methodological caveat on the MAD scale. The robust 3σ A4 detector uses $\hat{\sigma} = 1.4826 \cdot \text{MAD}$ as a Gaussian-consistent scale estimator. This assumes an approximately unimodal, isotropic spatial distribution of stations per system. For systems whose deployment is intrinsically anisotropic (transit corridors, riverside linear networks) the MAD over-estimates the effective dispersion on the dominant axis and the detector *under*-flags peripheral stations on that axis; for *multi-modal* deployments (nationwide systems clustering around many city centres; see the *Call a Bike* case in Table 10) the single-centroid construction *over*-flags. The threshold sweep above measures the sensitivity of the detector to its own σ parameter; it does not measure the sensitivity to the underlying geometric assumption, which is a known limitation and the subject of the sub-system-clustering refinement noted in Section 5.

S1 — A4 outlier-detector sensitivity to σ.

S2 — Empirical distribution of the A6 zero-capacity rate. The zero-capacity rate r_s^{A6} across the 301 docked systems with at least 20 stations is reported in Table S2; it underpins the $\tau_{A6} = 1\,\%$ threshold of Section 2.3.

Table S2: Empirical distribution of the zero-capacity rate r_s^{A6} across the 301 docked systems with at least 20 stations and a reachable `station_information` feed. Free-floating systems, where $c = 0$ is uninformative, are excluded from the A6 base.

Bucket	Systems	Share
$r_s^{A6} = 0\,\%$ (no $c = 0$)	268	89.0 %
$0 < r_s^{A6} \leq 0.5\,\%$	7	2.3 %
$0.5 < r_s^{A6} \leq 1\,\%$	4	1.3 %
$1 < r_s^{A6} \leq 2\,\%$	10	3.3 %
$2 < r_s^{A6} \leq 5\,\%$	8	2.7 %
$5 < r_s^{A6} \leq 10\,\%$	2	0.7 %
$r_s^{A6} > 10\,\%$	2	0.7 %
Summary statistics		
mean $\bar{r}$	0.36 %	
standard deviation $\hat{\sigma}$	2.41 %	
systems with $r_s^{A6} \geq \tau_{A6} = 1\,\%$	22	

S3 — Empirical distribution of the A7 null-capacity rate. The NaN-capacity rate ρ_s^{A7} across the 640 globally audited systems with at least 20 stations is reported in Table S3; its sharp bimodality justifies the coarse $\tau_{A7} = 0.5$ threshold of Section 2.4.

Table S3: Empirical distribution of the NaN-capacity rate ρ_s^{A7} across the 640 globally audited systems with at least 20 stations and a reachable `station_information` feed. The corpus is sharply bimodal at the extreme values, which justifies a coarse threshold rather than a distribution-anchored one.

Bucket	Systems	Share
$\rho_s^{A7} = 0\,\%$ (no NaN)	238	37.2 %
$0 < \rho_s^{A7} < 1\,\%$	4	0.6 %
$1 \leq \rho_s^{A7} < 10\,\%$	11	1.7 %
$10 \leq \rho_s^{A7} < 50\,\%$	22	3.4 %
$50 \leq \rho_s^{A7} < 99\,\%$	41	6.4 %
$99 \leq \rho_s^{A7} < 100\,\%$	5	0.8 %
$\rho_s^{A7} = 100\,\%$ (all NaN)	319	49.8 %
Mass at the two extremes	557	**87.0 %**

Table S4: Column families of `stations_gold_standard_final.parquet` (46 typed columns). Columns marked † are produced by the audit pipeline; * are derived from external sources. The full per-column schema (name, dtype, source, completeness) ships as a JSON Schema, a Frictionless Data Package and a Croissant manifest (`catalogue/metadata/`). The `audit_confidence` enum is *high* when no flag fires, *medium* for a single A3 or A7 flag, and *low* otherwise.

Family	Cols	Examples / source
Identifiers	5	`uid`, `station_id`, `system_id`
Spatial & administrative	6	`lat`/`lon`, `city`, INSEE commune/region
Station description	4	name, address, `capacity`, `n_stations_system`
Audit pipeline outputs†	13	`station_type`, `capacity_raw`/`_audited`, `flag_A1–A7`, `audit_confidence`
Network geometry†	5	intra-/inter-system KNN distances, local densities
Topography*	2	elevation, roughness (IGN BD ALTI)
Cycling infrastructure*	2	cycle-lane linear & share (IGN BD TOPO)
Safety*	1	severe cyclist crashes (ONISR BAAC)
Multimodal access*	2	heavy-transit stops within 300 m (GTFS)
Socio-economic context*	5	INSEE Filosofi income, Gini, car-less share
Modal share*	1	commute-by-bike share (INSEE)
Total	**46**	

Released catalogue schema (column families).

Reference sources for contextual enrichment.

Per-country breakdown of structural classes A1–A4.

Urban areas most affected by the purging protocol.

Empirical distribution of the A3 capacity-profile ratio.

A4 ablation: legacy versus topology-aware detector.

Table S5: Reference sources used for contextual enrichment. Every listed source contributes at least one column to the released parquet schema (Table S4). Out-of-scope sources considered during prototyping but not retained in the v1.0 release (OpenStreetMap point-of-interest density, FUB cycling barometer score) are documented in the audit report.

Source	Granularity	Variables	Operation
INSEE Filosofi	IRIS / municipality	Median income/CU, local Gini, decile, car-less share	Aggregation
INSEE Recensement	municipality	Share of commute by bike (`part_velo_travail`)	Aggregation
IGN BD ALTI	DEM raster	Elevation, topographic roughness index	Spatial sampl
IGN BD TOPO	polylines	Cycle-infrastructure linear (300 m buffer)	Spatial join
ONISR BAAC (Cerema)	geolocated accident	Severe-crash count (500 m, 5 yr)	Spatial join
National GTFS aggregator	GTFS stops	Heavy-transit stops within 300 m	Spatial join

Table S6: Per-country breakdown of the four structural classes A1–A4 on the MobilityData canonical catalogue (1,509 systems, including the 255 French entries). The macro-region class A5 (17 systems globally), zero-capacity A6 (22) and null-capacity A7 (245 / 97,314 stations) are calibrated on station-count-conditioned sub-corpora and are reported in Tables S2 and S3. Per-class counts are not disjoint; "Flagged" aggregates A1–A5 only. Countries with ≥ 30 audited systems or notable hotspot concentrations are shown. A4 requires both $> 5\,\%$ of stations and ≥ 5 absolute outside-country stations to flag, to avoid edge-of-coast false positives.

Country	Audited	Reach.	Flagged	A1	A2	A3	A4 geo
France (FR)	255	253	**38**	17	14	4	3
Germany (DE)	251	243	32	21	4	9	–
United States (US)	175	172	0	–	–	–	–
Poland (PL)	94	91	28	–	3	–	25
Norway (NO)	83	46	2	–	–	–	2
Spain (ES)	69	66	1	–	–	–	1
United Kingdom (GB)	56	54	3	–	1	–	2
Finland (FI)	55	55	14	–	–	–	14
Netherlands (NL)	52	52	1	–	–	–	1
Switzerland (CH)	49	46	15	8	3	3	1
Czech Rep. (CZ)	45	43	**25**	–	**16**	**12**	–
Belgium (BE)	37	34	4	–	–	2	2
Italy (IT)	34	33	0	–	–	–	–
Austria (AT)	27	23	1	–	1	1	–
Sweden (SE)	23	23	12	–	–	–	12
Slovakia (SK)	19	19	16	–	–	–	16
Total (48 countries)	1,509	1,420	204	46	48	33	81

Reproducibility. Per-country counts produced by `scripts/audit_live_systems.py --infer-type` against the 1,509-system MobilityData catalogue; the `--infer-type` flag activates operator-name heuristics (regexes for car-sharing / free-floating brands listed at the top of the script) required to fire A1 and A3 where `station_type` is not declared. Numbers re-fetched later will drift as feeds evolve; the 2026-04 snapshot used here is archived under `experiments/e2_threshold_sensitivity/global_audit_snapshot/` (run `verify_snapshot.py` to reproduce these counts). The sibling `global_audit_results_typed.csv` is a later 2026-05 re-crawl and does not match these figures.

Table S7: Top five French urban areas most affected by the purging protocol. “Raw” is the count of `station_information` entries in the unaudited feed; “Cert. dock” is the count of dock-based stations in the Audit Catalogue. The drop is essentially the share of GBFS entries that the audit reclassifies as free-floating (A3), car-sharing (A1), or filtered (A4–A5). The “Flags driving the drop” column lists the anomaly classes that each agglomeration’s flagged systems trigger.

Urban area	**Raw**	**Cert. dock**	**Drop**	**Flags driving the drop**
Bordeaux	9,921	225	$-97.7\,\%$	A3 (Pony), A1 (Citiz), A7 (Dott, Bird)
Paris	19,872	1,507	$-92.4\,\%$	A7 (Dott, Bird), A3 (Pony Paris)
Marseille	2,391	221	$-90.8\,\%$	A3 (Pony), A7 (Dott)
Strasbourg	310	39	$-87.4\,\%$	A1 (Citiz Grand Est)
Lyon	2,057	452	$-78.0\,\%$	A3 (Pony, Bird), A7 (Dott)

Table S8: Empirical distribution of $\bar{c}_{\text{profile}}/\bar{c}_{\text{actual}}$ across two corpora: the 95 audited French systems with non-empty `capacity` (split docked / free-floating; the derivation set) and the 267 globally audited systems (live sweep across 33 countries, station_type unknown for many; the out-of-sample set). The threshold $\tau_{A3} = 5$ is preserved in this paper as a conservative cut-off; the global distribution shows that the band $[2, 5]$ is not empty (15 systems) and that a future revision of A3 may benefit from a soft / hard two-tier threshold analogous to A6.

	French corpus (n = 95)		**Global (n = 267)**
Ratio bucket	**Docked**	**Free-floating**	**All**
$[1.00, 1.10]$	79	9†	197
$[1.10, 2.00]$	2	0	24
$[2.00, 5.00]$	0	0	15
$[5.00, 20.00]$	0	0	12
$[20.00, 50.00]$	0	0	9
$[50.00, \infty)$	0	5	10
Total	81	14	267

† Two qualitatively different cases populate the lowest bucket. The 79 docked French systems carry stable positive capacities so $\bar{c}_{\text{profile}}$ and $\bar{c}_{\text{actual}}$ coincide by virtue of *the data*; the 9 free-floating French systems sit there by virtue of *the arithmetic*, because their capacity column is mostly NaN or zero and the ratio degenerates to unity rather than reflecting the conditional-averaging publication choice. The same indistinguishability affects the global “All” column, where we cannot a-priori separate clean docks from degenerate-FF without a per-system audit; the top 10 global systems by ratio are however all known free-floating operators (Pony, nextbike, Voi, Bolt), so the $[5, \infty)$ tail tracks the A3 mechanism across the sweep.

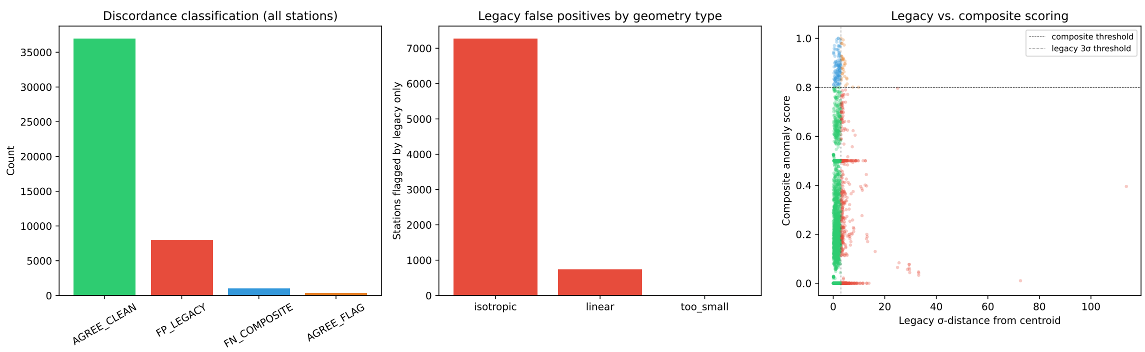

Figure S1: Ablation of the centroid-based A4 detector against the topology-aware composite detector. *Left*: discordance classification across all 46,307 stations. *Centre*: legacy false positives stratified by system geometry type (isotropic vs. linear). *Right*: scatter of legacy σ-distance against composite anomaly score, coloured by discordance class; dashed lines mark the respective detection thresholds.

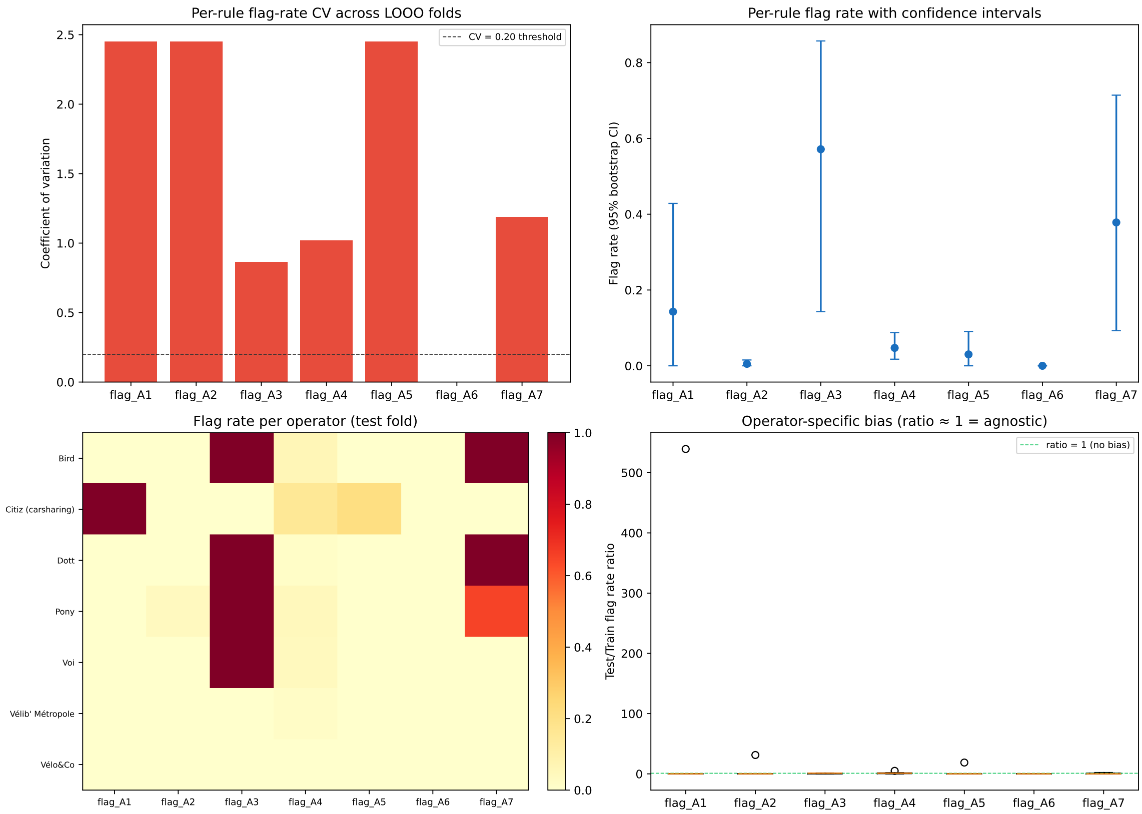

Figure S2: LOOO cross-validation overview. *Top-left*: per-rule coefficient of variation (green = CV < 0.20, red = above). *Top-right*: mean flag rate with 95 % bootstrap CI. *Bottom-left*: heatmap of per-operator flag rates on the test fold. *Bottom-right*: test/train flag-rate ratio (ratio ≈ 1 indicates operator-agnostic behaviour).

Leave-one-operator-out cross-validation overview.

Table S9: Mapping of the nine-step purging-protocol design properties to the data-quality dimensions of ISO/IEC 25012:2008 and its companion measurement standard ISO/IEC 25024:2015. The Audit Catalogue is, to our knowledge, the first FAIR-released implementation of these dimensions on a GBFS corpus.

Protocol property	ISO/IEC 25012 dimension	Mechanism in the pipeline
Idempotence	Internal consistency	Unit-tested re-application on certified output
Reversibility	Recoverability + Traceability	`rejected_stations.parquet` with exclusion motiv
Step-level logging	Traceability	Per-step in/out counts and transformation type
Explicit parameterisation	Understandability	Hyperparameters in code + documented in audit re
FAIR release	Compliance + Accessibility	JSON Schema, DCAT-AP, Croissant, ODbL on Zen
Blind author cross-annotation	Accuracy + Semantic consistency	Author-led consistency check that flagged anomalie

Mapping of the purging protocol to ISO/IEC 25012.

Urban areas most affected by the A3 reclassification.

Completeness of contextual enrichment.

Per-country incidence of structural classes A1–A4.

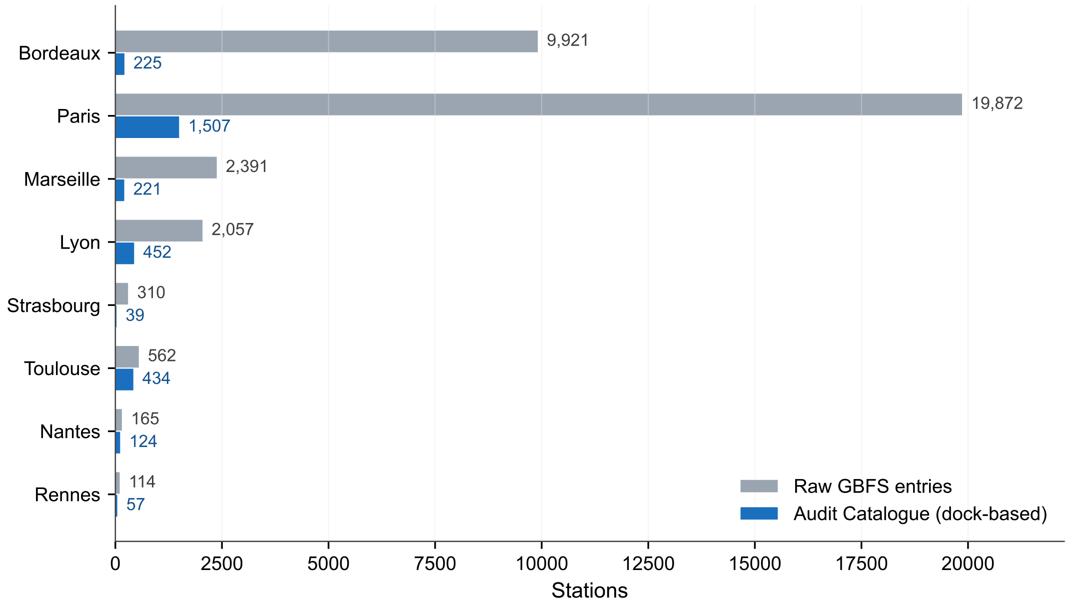

Figure S3: Top twelve French urban areas ranked by absolute reduction in dock-based station count between the raw GBFS feed and the certified Audit Catalogue. **Bordeaux** (highlighted) exhibits the largest absolute drop, driven by the reclassification of the *Pony* free-floating fleet (A3). Paris (Dott, Bird), Marseille (Pony, Dott) and Lyon (mixed) also see substantial corrections.

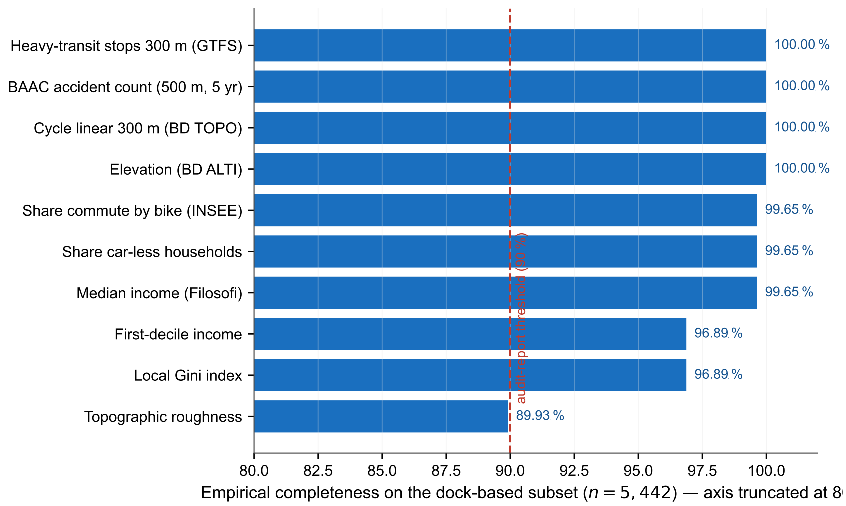

Figure S4: Empirical completeness of contextual enrichment per derived variable on the dock-based subset of the Audit Catalogue ($n = 5{,}442$). All variables exceed the 90 % completeness threshold required by the audit report; missing values remain below 6 % across the corpus.

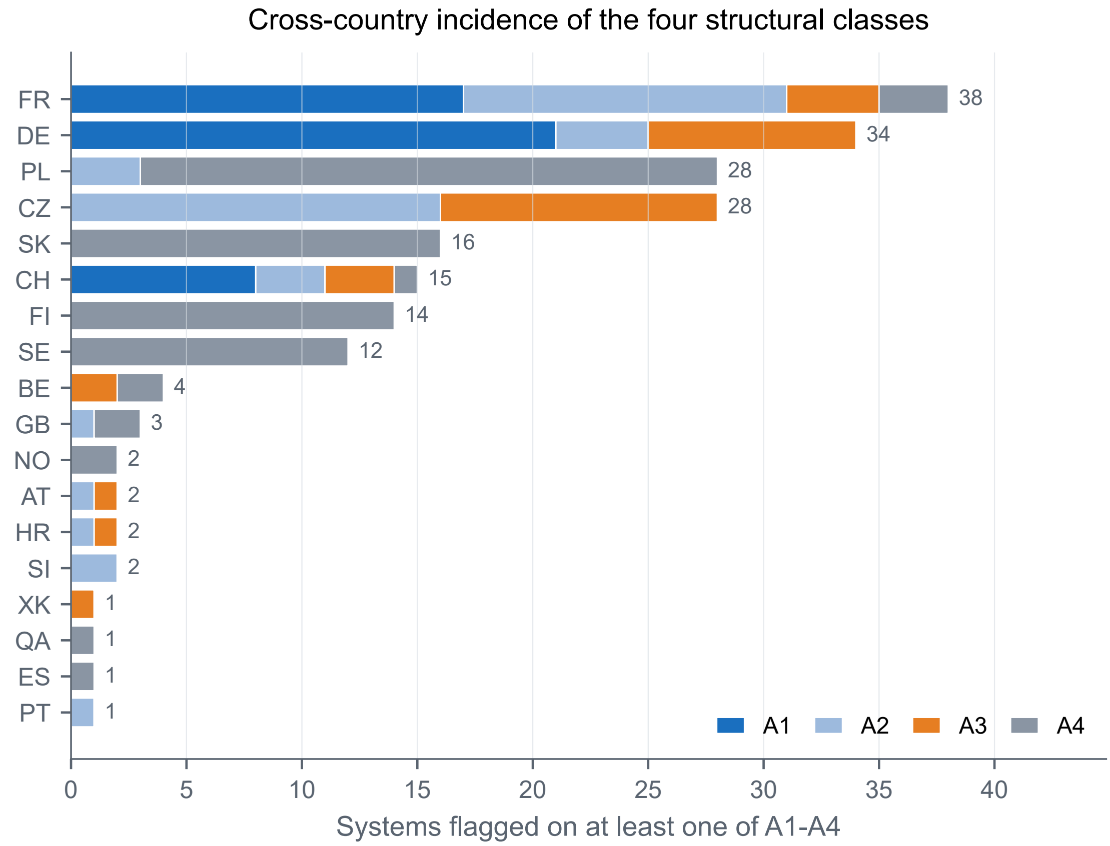

Figure S5: Per-country incidence of the structural classes A1–A4 across the 1,509-system MobilityData catalogue (top 18 countries by total flagged). The chart shows the stacked composition of the **Flagged** column of Table S6 and makes the country-level signatures legible at a glance: A4 (geospatial) drives the Polish, Finnish, Swedish and Slovak hotspots (cross-border bounding boxes on aggregator feeds), A1 (car-sharing labelled as BSS) drives the DACH cluster (Germany, Switzerland), A2 (placeholder capacity) and A3 (over-capacity) drive France and the Czech Republic. Macro-region (A5), zero-capacity (A6) and null-capacity (A7) are reported separately because they are calibrated on station-count-conditioned sub-corpora (Tables S2 and S3).

Table S10: Nine falsifiable experiments mapped to the limitations of Section 5.5. "✓" = completed result reported in this paper; "Pilot ✓" = preliminary result already reported; full protocols and reproduction scripts ship under `experiments/` in the source repository.

ID	Title	Targets	Status
E1	Pre-registered held-out audit (post-freeze MobilityData)	Construct validity / circularity	Retrospective version
E2	Threshold and buffer sensitivity	$\sigma_{\max}$, $N_{\min}$, radii	$\sigma_{\max}$ pilot ✓
E3	Twelve-month temporal stability	Snapshot dependence (L4)	Running
E4	Dynamic extension to `station_status`	L2	2-day pilot ✓; Shanno
E5	European generalisation (live audit, 6 tech stacks)	L1	13-system panel ✓
E6	Free-floating-native protocol	L3	Planned 2027-Q2
E7	**A4 v2 topology-aware ablation**	**Geospatial isotropy bias**	✓ **(Sec. 4.2)**
E8	**Leave-one-operator-out CV**	**Overfitting (H3)**	✓ **(Sec. 4.3)**
E9	**Dynamic station_status audit**	**Static/dynamic gap (L2)**	**Future work (pipel**

Nine falsifiable experiments mapped to the limitations.